EPGSpeller: Towards Fast, Hands-Free AAC Text Entry with Electropalatographic Silent Spelling

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1 Introduction

Hands-free text entry is a central problem for augmentative and alternative communication (AAC): many users and usage settings cannot assume reliable speech, hand movement, or a visible keyboard. Silent speech interfaces offer an appealing route, but a practical AAC channel must support more than a small set of pre-defined commands. It must let users compose words and messages.

Electropalatography (EPG) makes this setting concrete. A SmartPalate-style sensor records tongue-palate contact as a multichannel temporal pattern while the user silently articulates. In the SilentSpeller-style recordings used in this work, each trial is a variable-length EPG sequence paired with the word that the participant attempted to spell or articulate.

Prior silent-speech interfaces have shown that silent articulatory or biosignal patterns can be mapped to useful commands or constrained text-entry actions. That capability is important, but it is not the same as an AAC text channel that lets a user spell arbitrary messages from character evidence.

The key interface question is therefore not simply whether EPG can recognize a known silent command. EPGSpeller asks whether the contact signal can be decoded at the character level. Character-level silent spelling breaks the fixed-vocabulary limitation of command recognition: if the spelling stream is recognized well enough, the user is no longer restricted to words that were present as output classes during training. The related-work comparison also motivates keeping command recognition, modality-specific SSI systems, and LM-assisted text entry separate from the present lexicon-free EPG spelling claim.

This paper presents EPGSpeller, a character-level recognizer for electropalatographic silent spelling. The contribution is not only a lower-error decoder. The system shows that EPG can be framed as a hands-free AAC text-entry interface that is no longer tied to a fixed command vocabulary. At the same time, the experiments deliberately expose the remaining bottleneck: personalized calibration is effective, whereas cross-participant transfer and high-shot calibration are still constrained by the available data. The preprocessing sweep reinforces this interpretation by showing that simple normalization changes cross-user errors but does not turn them into personalized recognition.

The evidence is organized around that claim boundary. The paper first tests personalized and word-holdout spelling recognition, then asks whether calibration can be reduced, and finally checks whether multi-source transfer or simple preprocessing removes the user-dependence.

The contributions are: (1) an AAC-centered framing of SmartPalate EPG as a character-level silent spelling channel, (2) a high-performing lexicon-free vector/GRU recognizer for that task, and (3) evidence that connects personalized spelling, SilentSpeller-compatible word holdout, and calibration limits without hiding the remaining user-dependence behind a single benchmark score.

2 Task and Method

The input representation is the EPG contact stream itself: each trial is treated as an ordered sequence of palate-contact frames rather than as audio, video, or a manually segmented phone sequence. This keeps the task tied to the SmartPalate-style signal and makes the decoding claim about silent tongue-palate contact.

Given an EPG contact sequence, the model outputs a character string. We treat greedy character-level decoding as the primary lexicon-free result. Lexicon-projected scores are useful diagnostics, but they are not allowed to carry the central claim because an AAC spelling interface must eventually support words and messages outside a fixed evaluation lexicon.

EPGSpeller uses a character-level CTC recognizer over EPG frames. We evaluate a high-capacity vector/GRU recognizer as the main system and compare it with layout-aware variants that use row-column pooling or two-dimensional palatal grids. The design goal is deliberately pragmatic: maximize silent spelling accuracy under lexicon-free decoding, then use ablations to ask which EPG-specific inductive biases help under the same protocols.

Table 1: Dataset audit summary for the SilentSpeller-style EPG recordings.

id	N	V	Tmed	contact	zero
p1	2328	1164	234	0.19	0
p2	2328	1164	196	0.20	0
p3	2797	1164	283	0.12	4
p4	1167	1162	229	0.19	0

Table 2: Main protocol recap for the vector/GRU recognizer. CER is greedy character error rate; lex_all is lexicon-projected error.

protocol	n	cer	lex
P1	4	0.180±0.084	0.102±0.068
P2	3	0.145±0.063	0.075±0.048
P3	6	0.691±0.133	0.644±0.060

The method scope is therefore intentionally narrow. The vector/GRU system is the main recognizer used for the paper's interface claim, while layout-aware and reduced-electrode variants are treated as controlled engineering ablations under the same decoding and reporting rules.

3 Evaluation

The evaluation uses four SilentSpeller-style EPG datasets with different repetition densities: p1 and p2 each contain 2328 trials over 1164 word types, p3 contains 2797 trials over 1164 word types, and p4 contains 1167 trials over 1162 word types. The dataset summary table supports the feasibility decisions used later in the calibration analysis, because low-shot and high-shot calibration depend on repeated productions per word, not only on the number of participants.

The evaluation separates the claims that matter for an AAC spelling interface. Personalized protocols measure whether the recognizer can work after user-specific training; word-holdout protocols ask whether spelling generalizes beyond training word types; cross-participant protocols test whether a model can be reused without target calibration; and k-shot protocols measure how much target-user calibration is needed. Reported tables aggregate repeated runs by condition and keep incompatible protocols separate rather than pooling instance holdout, word holdout, and cross-participant transfer into one score.

The primary metric is greedy character error rate (CER), computed without projecting the output onto a candidate lexicon. Where useful, we also report lexicon-projected scores to compare with closed-vocabulary behavior, but those values are not used as the main evidence for lexicon-free text entry. Real-time factor (RTF) is reported separately to show that decoding latency is not the main obstacle in the present experiments.

Baselines and controls are separated by what they test. The main vector/GRU path carries the recognition result; SilentSpeller-compatible-lite word holdout connects to the historical task; lexicon projection is reported only as a closed-vocabulary diagnostic; and preprocessing, spatial-front-end, and electrode-budget variants are used as ablations rather than as alternate definitions of the primary task. This separation is what keeps the lexicon-free claim attached to greedy CER instead of to a lexicon projection or a favorable post-processing condition.

4 Results

Table 2 gives the central recognition result. With the same vector/GRU recognizer and greedy decoding, personalized conditions are much stronger than cross-participant transfer: P1 reaches 0.1796 CER, P2 reaches 0.1451 CER, whereas P3 cross-participant transfer remains at 0.6909 CER. The contrast is the main empirical shape of the paper: EPG silent spelling is viable after user-specific calibration, but it is not yet a calibration-free interface.

To connect this framing back to SilentSpeller-style evaluation, we additionally use a word-holdout lite condition on subj1 with 100 unseen test word types. The vector/GRU recognizer obtains 0.1133 +/- 0.0128 greedy CER across four seeds. This is not claimed as a full recreation of the historical implementation; it is a same-dataset, spelling-oriented bridge showing that the recognizer remains effective when the test words are held out.

The calibration experiments turn the cross-user weakness into a more actionable result. For subj3, zero-shot transfer gives 0.8403 CER for single-source transfer and 0.8381 CER for multi-source transfer, while one and two target examples per word reduce CER to 0.3503 and 0.2458. Additional k=1 runs on subj1 and subj2 give 0.1082 and 0.1118 CER, indicating that one-shot personalized calibration can be strong when each word has two productions. However, the available public data do not

Table 3: SilentSpeller-compatible-lite word-holdout result for subj1.

condition	n_test_words	n	cer_m	cer_s	rtf_m	rtf_s
subj1_test100	100	4	0.113343	0.012833	0.000290	0.000042

Table 4: Calibration curve and feasibility for subj3. The k=4 row is reduced-vocabulary diagnostic evidence; k=8 is data-infeasible.

target	k	condition	n	cer_m	cer_s	rtf_m	rtf_s	notes
subj3	0	P3_single_source	8	0.840301	0.094775	0.000097	0.000004	msx/uc vector
subj3	0	P3MS_multi_source	4	0.838062	0.033612	0.000096	0.000002	msx:p3ms vector
subj3	1	P2K_within_subject	4	0.350280	0.059950	0.000164	0.000009	msx:existing
subj3	2	P2K_within_subject	4	0.245802	0.008507	0.000165	0.000002	msx:existing
subj3	4	P2K_reduced_vocab	4	0.781609	0.208381	0.002482	0.000169	sltp2:new reduced vocabulary (min_repetitions=
subj3	8	N/A	0					infeasible (feasible_vocab=0, feasible_for_50_wo

support a standard 50-word held-out evaluation at k=4 or k=8: only 14 word types satisfy the k=4 repetition requirement for subj3, and none satisfy k=8.

Multi-source transfer does not remove the bottleneck. Paired-source and multi-source variants sometimes improve over a single-source baseline, but the resulting errors are still far from personalized performance and vary strongly by target. For target subject 4, all-to-4 vector CER is 0.686 in P3 and 0.582 in P3MS. A preprocessing sweep over raw, standardized, contact-matched, target-marginal, and PCA-standardized variants improves the best protocol-level averages only to 0.6426 for P3 and 0.5692 for P3MS. For the paper's AAC framing, this means the current evidence supports personalized silent spelling, not an off-the-shelf model that can be deployed to a new user without calibration.

These transfer results are interpreted alongside validation controls, not as an isolated negative table. Multi-source training, target-specific transfer directions, and preprocessing variants all test whether the observed gap can be attributed to a simple protocol artifact. They reduce some ambiguities, but they do not change the claim boundary: the present system is supported as personalized lexicon-free spelling, not as calibration-free deployment.

The strongest limitation is not decoding speed or the existence of a recognizer. It is calibration. The experiments show that EPGSpeller can work well when participant-specific data are available, but cross-user transfer remains weak. The preprocessing ablation rules out a simple explanation in which train-only standardization, contact-rate matching, target marginal normalization, or PCA alone closes the gap; the best protocol-level preprocessing averages remain 0.6426 CER for P3 and 0.5692 CER for P3MS. The experiments also show that the present public recordings do not contain enough repeated productions per word to run a standard k=4 or k=8 50-word calibration curve. Future AAC-oriented EPG datasets therefore need denser repeated spelling trials, not only more model variants.

5 Analysis

The preprocessing sweep tests whether the cross-user result is just a scale or contact-rate artifact. It covers 120 runs over six transfer directions, four seeds, and five preprocessing conditions: raw standardized, raw unstandardized, contact-rate matched, target-marginal normalized, and PCA-standardized. The best protocol-level averages are still 0.6426 CER for P3 and 0.5692 CER for P3MS. This supports the calibration-bottleneck interpretation: simple normalization helps in some directions but does not create a calibration-free EPG spelling system.

The electrode-budget analysis is best used as a practical robustness result. A reduced 64-electrode representation can preserve much of the performance, but within-subject top-k, farthest-point selection, transfer-derived selection, and random subsets are close enough that the paper should not claim a universal channel-selection rule.

The spatial ablations are important but should not become the paper's headline. Row-column pooling is often close to the vector baseline, whereas two-dimensional grid variants can increase CER and RTF. The result is useful because it prevents an overly simple story that palatal layout always helps; in this dataset, the fastest and most reliable spelling recognizer is the vector/GRU baseline.

6 Auditability

The generated TeX is a reproducible draft view, not the only editable manuscript. Venue-specific cleanup can be copied into paper/final while preserving qpaper block anchors, so later checks can still relate final prose back to the question cards and

Table 5: Low-shot extension beyond subj3 for subjects with two repetitions per word.

subject	k	n	cer_m	cer_s	rtf_m	rtf_s	source
subj1	1	4	0.108222	0.005059	0.000106	0.000001	sltp2:new
subj2	1	4	0.111769	0.006698	0.000117	0.000001	sltp2:new
subj3	1	4	0.350280	0.059950	0.000164	0.000009	msx:existing
subj3	2	4	0.245802	0.008507	0.000165	0.000002	msx:existing

Table 6: Multi-source cross-participant transfer results.

group	variant	cer	rtf
p23-¿p1	vec	0.473±0.014	0.0001±0.0000
p23-¿p1	rowcol	0.476±0.014	0.0002±0.0000
p23-¿p1	grid	0.602±0.076	0.0004±0.0000
p23-¿p1	grid_aug	0.691±0.165	0.0004±0.0000
p13-¿p2	vec	0.520±0.003	0.0001±0.0000
p13-¿p2	rowcol	0.504±0.012	0.0003±0.0000
p13-¿p2	grid	0.511±0.047	0.0004±0.0000
p13-¿p2	grid_aug	0.511±0.016	0.0004±0.0000
p12-¿p3	vec	0.838±0.034	0.0001±0.0000
p12-¿p3	rowcol	0.790±0.033	0.0002±0.0000
p12-¿p3	grid	0.736±0.018	0.0003±0.0000
p12-¿p3	grid_aug	0.756±0.020	0.0003±0.0000

evidence registry.

For review and revision, the qpaper source keeps the manuscript organized as claim-bearing question cards. Each result paragraph and table points to an observed artifact ID in evidence/index.yaml, and the generated trace files expose the mapping from claims to evidence. This audit layer is not the scientific contribution, but it reduces the risk that paper text, generated tables, and experiment artifacts drift apart during submission editing.

Table 7: Additional cross-participant generalization results for target subject 4.

proto	lvl	group	cer	lex
P3	dir	p1-¿p4	0.666±0.024	0.651±0.026
P3	dir	p2-¿p4	0.809±0.021	0.736±0.022
P3	dir	p3-¿p4	0.584±0.018	0.591±0.029
P3	all	all-¿p4	0.686±0.114	0.659±0.073
P3MS	dir	p12-¿p4	0.655±0.016	0.593±0.016
P3MS	dir	p13-¿p4	0.536±0.009	0.533±0.008
P3MS	dir	p23-¿p4	0.555±0.005	0.537±0.011
P3MS	all	all-¿p4	0.582±0.064	0.554±0.034

Table 8: Preprocessing ablation summarized by protocol and condition.

protocol	condition	n	cer_m	cer_s	lex_all_m	lex_all_s	rtf_m	rtf_s
P3	raw_std	12	0.6893	0.0951	0.6632	0.0661	0.000117	0.000007
P3	raw_none	12	0.6527	0.0545	0.6351	0.0352	0.000122	0.000012
P3	raw_contact_match	12	0.6773	0.0973	0.6533	0.0661	0.000121	0.000009
P3	raw_target_marginal	12	0.6426	0.0811	0.6104	0.0508	0.000116	0.000002
P3	pca32_std	12	0.6960	0.0836	0.6682	0.0580	0.000116	0.000004
P3MS	raw_std	12	0.5818	0.0535	0.5552	0.0307	0.000137	0.000038
P3MS	raw_none	12	0.5848	0.0304	0.5637	0.0186	0.000120	0.000006
P3MS	raw_contact_match	12	0.5747	0.0657	0.5433	0.0389	0.000125	0.000016
P3MS	raw_target_marginal	12	0.5692	0.0544	0.5310	0.0322	0.000126	0.000015
P3MS	pca32_std	12	0.5919	0.0523	0.5658	0.0284	0.000125	0.000030

Table 9: Preprocessing ablation by transfer direction.

protocol	direction	condition	n	cer_m	cer_s	lex_all_m	lex_all_s	rtf_m	rtf_s
P3MS	subj12to4	raw_std	4	0.6489	0.0166	0.5907	0.0102	0.000140	0.000041
P3MS	subj12to4	raw_none	4	0.6158	0.0177	0.5603	0.0072	0.000120	0.000005
P3MS	subj12to4	raw_contact_match	4	0.6579	0.0195	0.5874	0.0100	0.000128	0.000018
P3MS	subj12to4	raw_target_marginal	4	0.6394	0.0234	0.5698	0.0111	0.000119	0.000003
P3MS	subj12to4	pca32_std	4	0.6578	0.0137	0.6002	0.0108	0.000141	0.000052
P3MS	subj13to4	raw_std	4	0.5320	0.0196	0.5237	0.0182	0.000131	0.000027
P3MS	subj13to4	raw_none	4	0.5505	0.0084	0.5518	0.0181	0.000116	0.000003
P3MS	subj13to4	raw_contact_match	4	0.5097	0.0052	0.5012	0.0141	0.000126	0.000023
P3MS	subj13to4	raw_target_marginal	4	0.5239	0.0103	0.5011	0.0164	0.000139	0.000021
P3MS	subj13to4	pca32_std	4	0.5398	0.0028	0.5393	0.0102	0.000115	0.000003
P3	subj1to4	raw_std	4	0.6877	0.0220	0.6708	0.0367	0.000119	0.000011
P3	subj1to4	raw_none	4	0.6221	0.0207	0.5977	0.0149	0.000120	0.000009
P3	subj1to4	raw_contact_match	4	0.6705	0.0196	0.6648	0.0257	0.000122	0.000005
P3	subj1to4	raw_target_marginal	4	0.6174	0.0017	0.5860	0.0110	0.000115	0.000001
P3	subj1to4	pca32_std	4	0.6818	0.0099	0.6621	0.0169	0.000113	0.000004
P3MS	subj23to4	raw_std	4	0.5646	0.0116	0.5511	0.0021	0.000141	0.000053
P3MS	subj23to4	raw_none	4	0.5882	0.0124	0.5791	0.0193	0.000123	0.000008
P3MS	subj23to4	raw_contact_match	4	0.5564	0.0100	0.5413	0.0166	0.000121	0.000006
P3MS	subj23to4	raw_target_marginal	4	0.5444	0.0086	0.5222	0.0104	0.000120	0.000001
P3MS	subj23to4	pca32_std	4	0.5782	0.0135	0.5580	0.0117	0.000119	0.000007
P3	subj2to4	raw_std	4	0.8003	0.0119	0.7316	0.0173	0.000117	0.000005
P3	subj2to4	raw_none	4	0.7209	0.0314	0.6717	0.0065	0.000117	0.000003
P3	subj2to4	raw_contact_match	4	0.7911	0.0419	0.7216	0.0153	0.000116	0.000003
P3	subj2to4	raw_target_marginal	4	0.7467	0.0209	0.6773	0.0035	0.000116	0.000002
P3	subj2to4	pca32_std	4	0.7993	0.0195	0.7364	0.0106	0.000116	0.000004
P3	subj3to4	raw_std	4	0.5799	0.0114	0.5872	0.0190	0.000114	0.000003
P3	subj3to4	raw_none	4	0.6151	0.0125	0.6358	0.0249	0.000130	0.000018
P3	subj3to4	raw_contact_match	4	0.5704	0.0038	0.5734	0.0143	0.000126	0.000015
P3	subj3to4	raw_target_marginal	4	0.5638	0.0105	0.5680	0.0120	0.000117	0.000002
P3	subj3to4	pca32_std	4	0.6071	0.0118	0.6061	0.0230	0.000117	0.000005

Table 10: Comparison of 64-electrode selection methods.

protocol	method	n	cer_m	cer_s	rtf_m	rtf_s	variant_id
P1	within_topk	4	0.1952	0.0827	0.000657	0.000210	uni_gru_within_topk64
P1	within_fps2k	4	0.1912	0.0781	0.000619	0.000229	uni_gru_within_fps2k64
P1	transfer_topk	4	0.2053	0.0951	0.000586	0.000218	uni_gru_transfer_subj1_topk64
P1	random	4	0.1980	0.0793	0.000665	0.000282	uni_gru_random64_seed20260224
P2	within_topk	3	0.1544	0.0591	0.000110	0.000011	uni_gru_within_topk64
P2	within_fps2k	3	0.1539	0.0617	0.000110	0.000010	uni_gru_within_fps2k64
P2	transfer_topk	3	0.1584	0.0709	0.000106	0.000009	uni_gru_transfer_subj1_topk64
P2	random	3	0.1566	0.0669	0.000105	0.000011	uni_gru_random64_seed20260224
P3	within_topk	6	0.6473	0.1301	0.000108	0.000009	uni_gru_within_topk64
P3	within_fps2k	6	0.6564	0.1465	0.000107	0.000010	uni_gru_within_fps2k64
P3	transfer_topk	6	0.6569	0.1441	0.000109	0.000009	uni_gru_transfer_subj1_topk64
P3	random	6	0.6442	0.1328	0.000111	0.000010	uni_gru_random64_seed20260224

Table 11: Spatial front-end comparison with the full electrode set.

protocol	variant	n	cer_m	cer_s	rtf_m	rtf_s	variant_id
P1	vector	4	0.1796	0.0844	0.000628	0.000232	uni_gru
P1	rowcol	4	0.2016	0.0886	0.002401	0.001106	rowcol_uni_gru
P1	spatial2d	4	0.3062	0.1491	0.003564	0.001540	spatial2d_uni_gru_aug0
P1	spatial2d_aug	4	0.3094	0.1440	0.003414	0.001054	spatial2d_uni_gru_aug1
P2	vector	3	0.1451	0.0627	0.000107	0.000010	uni_gru
P2	rowcol	3	0.1613	0.0783	0.000244	0.000037	rowcol_uni_gru
P2	spatial2d	3	0.2894	0.2126	0.000389	0.000047	spatial2d_uni_gru_aug0
P2	spatial2d_aug	3	0.3139	0.2436	0.000392	0.000045	spatial2d_uni_gru_aug1
P3	vector	6	0.6909	0.1332	0.000110	0.000014	uni_gru
P3	rowcol	6	0.6828	0.1510	0.000243	0.000033	rowcol_uni_gru
P3	spatial2d	6	0.7507	0.0897	0.000393	0.000042	spatial2d_uni_gru_aug0
P3	spatial2d_aug	6	0.7559	0.0920	0.000397	0.000046	spatial2d_uni_gru_aug1

Table 12: Spatial front-end comparison at a 64-electrode budget.

protocol	variant	n	cer_m	cer_s	rtf_m	rtf_s	variant_id
P1	vector	4	0.1952	0.0827	0.000657	0.000210	uni_gru_within_topk64
P1	rowcol	4	0.2005	0.0847	0.002487	0.001145	rowcol_uni_gru_within_topk64
P1	spatial2d	4	0.3796	0.2416	0.003103	0.001343	spatial2d_uni_gru_aug0_within_topk64
P1	spatial2d_aug	4	0.3657	0.2070	0.003117	0.001337	spatial2d_uni_gru_aug1_within_topk64
P2	vector	3	0.1544	0.0591	0.000110	0.000011	uni_gru_within_topk64
P2	rowcol	3	0.1617	0.0702	0.000242	0.000038	rowcol_uni_gru_within_topk64
P2	spatial2d	3	0.3882	0.3314	0.000389	0.000051	spatial2d_uni_gru_aug0_within_topk64
P2	spatial2d_aug	3	0.3810	0.3208	0.000391	0.000052	spatial2d_uni_gru_aug1_within_topk64
P3	vector	6	0.6473	0.1301	0.000108	0.000009	uni_gru_within_topk64
P3	rowcol	6	0.6611	0.1486	0.000256	0.000027	rowcol_uni_gru_within_topk64
P3	spatial2d	6	0.7470	0.1306	0.000405	0.000037	spatial2d_uni_gru_aug0_within_topk64
P3	spatial2d_aug	6	0.7535	0.1244	0.000392	0.000048	spatial2d_uni_gru_aug1_within_topk64